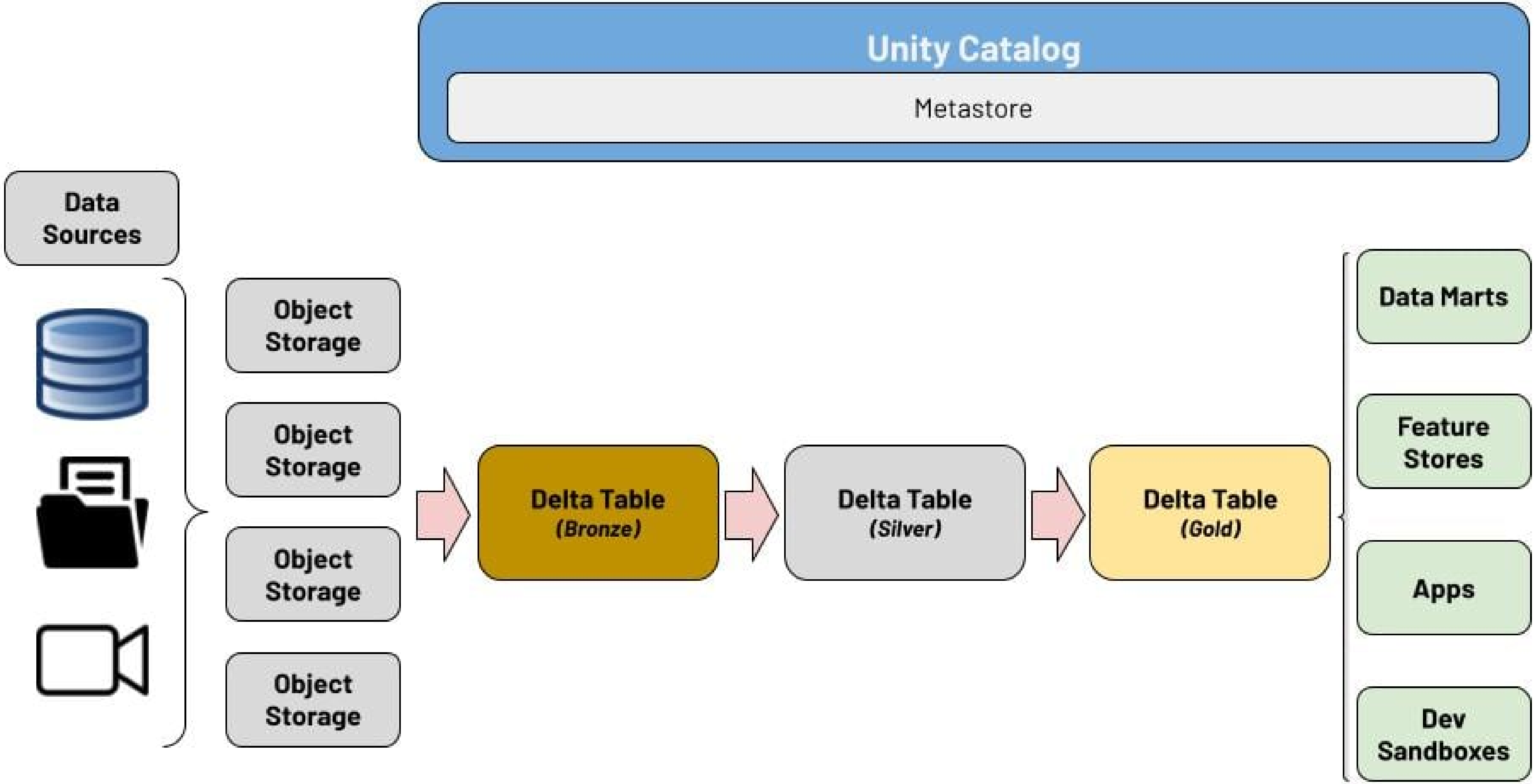
Common data engineering patterns

INTRODUCTION TO DATABRICKS SQL

Kevin Barlow
Data Manager

Motivation

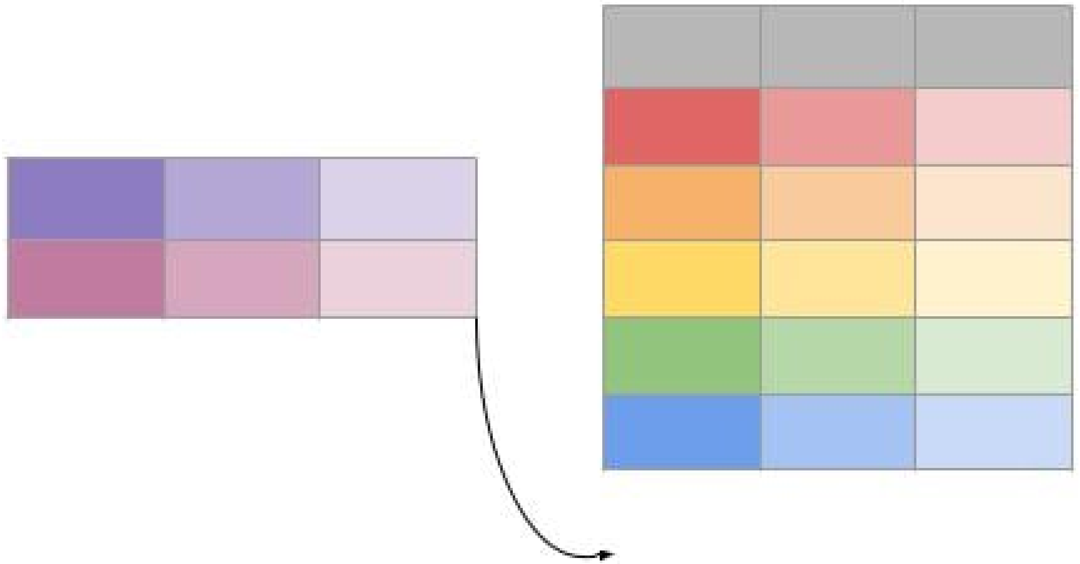

Unity Catalog
Metastore
Data Sources
Object Storage
Object Storage
Object Storage
Object Storage
Delta Table (Bronze)
Delta Table (Silver)
Delta Table (Gold)
Data Marts
Feature Stores
Apps
Dev Sandboxes

Handling incoming data

Incremental append

- Add all new data to end of existing table

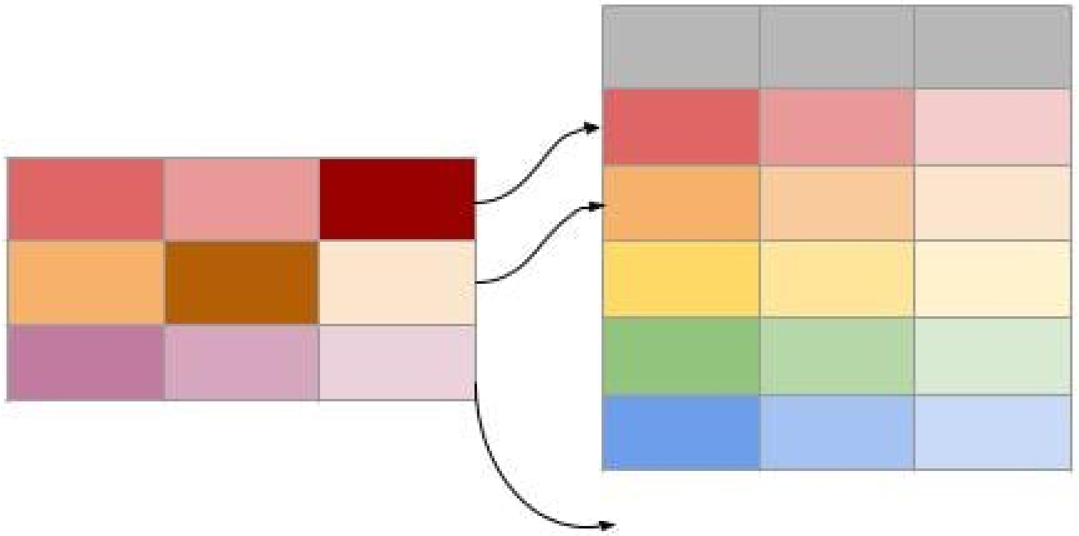
```
INSERT INTO students
TABLE visiting_students;
```



Change Data Capture (CDC)

- Integrates data into existing table

```
MERGE INTO target USING source
ON target.key = source.key
WHEN MATCHED THEN UPDATE SET *;
```



Data optimizations

OPTIMIZE

- Compact subset of data together
- Reduces the "small file problem"

Z-ORDER

- Similar to indexing for database systems
- Co-locates related data to same files
- Can reduce time to read data

```
> OPTIMIZE table_name;

> OPTIMIZE table_name
    WHERE date >= '2024-01-01';

> OPTIMIZE table_name
    WHERE date >= current_timestamp()
        - INTERVAL 1 day
    ZORDER BY (eventType);
```

Let's practice!

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Updating coffee sales data

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Advanced data analysis patterns

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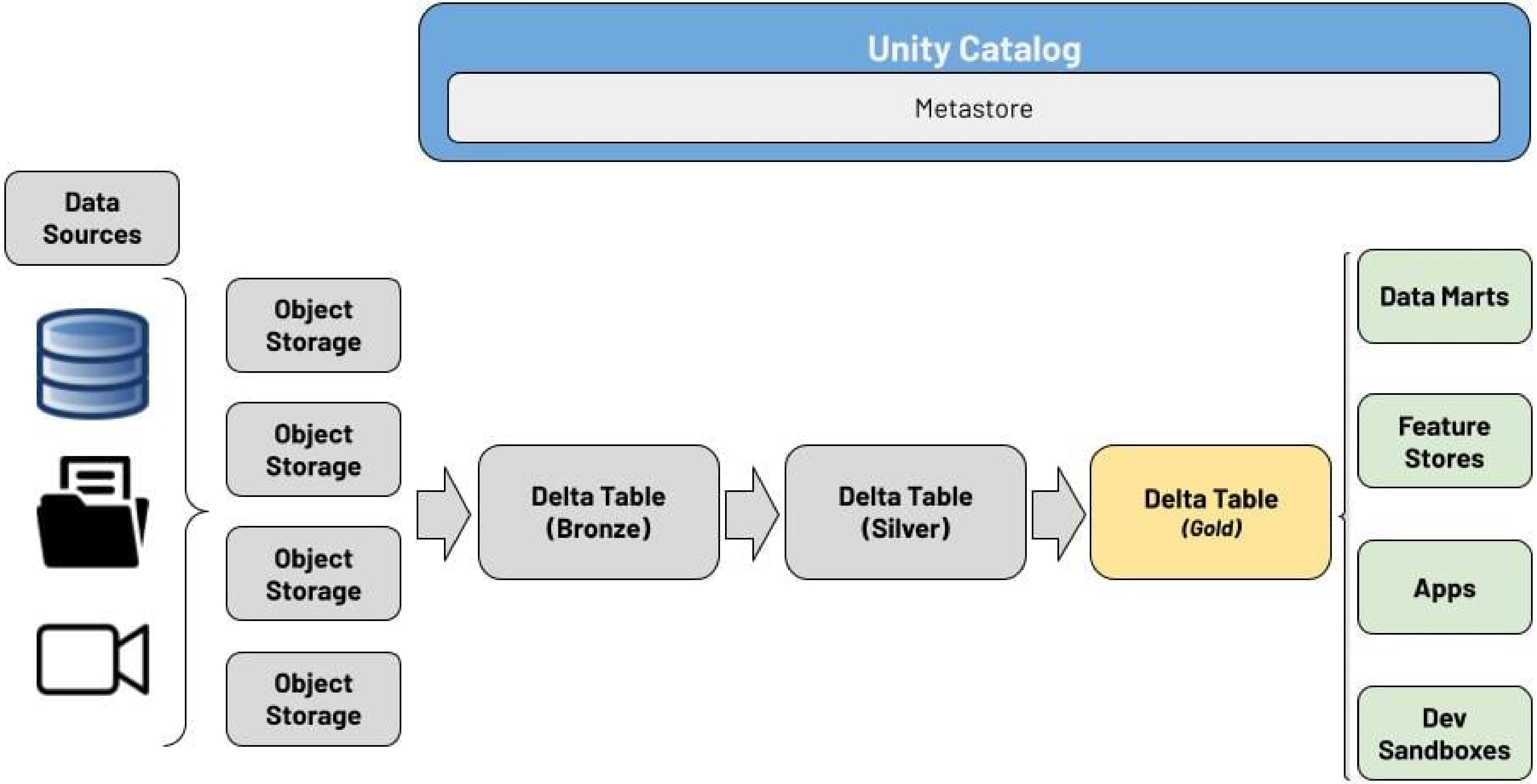

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Sub queries

- Achieved through "nesting" SQL queries
  - Nested query is thought of as a new "table"
  - Effectively similar to creating a view
- Great for a variety of use cases
  - Retrieve results of another query
  - Simplify complex queries
  - Aggregation within a query

```
SELECT store, totalRev, product
```

```
FROM (
    SELECT count(*) as count,
           sum(revenue) as totalRev,
           min(price) as minPrice,
           max(units) as maxUnits,
           region,
           store,
           product
      FROM sales
      GROUP BY region, store, product
) q
```

Window functions

- A category of SQL functions and techniques
- Performs a calculation across a specific range of rows
  - Rows have some kind of relation to each other
- Examples:
  - Calculating a metric based on a time range
  - Calculating change over subsequent rows
  - Calculating across several dimensions

```
SELECT name,
       dept,
       RANK() OVER
          (PARTITION BY dept
        ORDER BY salary) AS rank
  FROM employees;
```

name	dept	salary	rank
Lisa	Sales	10000	1
Alex	Sales	30000	2
Fred	Engineering	21000	1
Tom	Engineering	23000	2

Advanced Databricks SQL functions

RANK()

- Compares rows within a given partition and calculated a rank for each row

```
SELECT a,
       b,
       RANK() OVER(PARTITION BY a ORDER BY b DESC),
    FROM table_name;
```

a	b	rank
A1	3	1
A1	1	3
A1	2	2
A2	1	1

LAG() and LEAD()

- Returns either the preceding (`LAG()`) or subsequent (`LEAD()`) value from row

```
SELECT a,
       b,
       LAG(b) OVER (PARTITION BY a ORDER BY b)
    FROM table_name;
```

a	b	lag
A1	3	NULL
A1	1	3
A1	2	1
A2	1	NULL

Let's practice!

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Analyzing coffee sales by store

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Course recap

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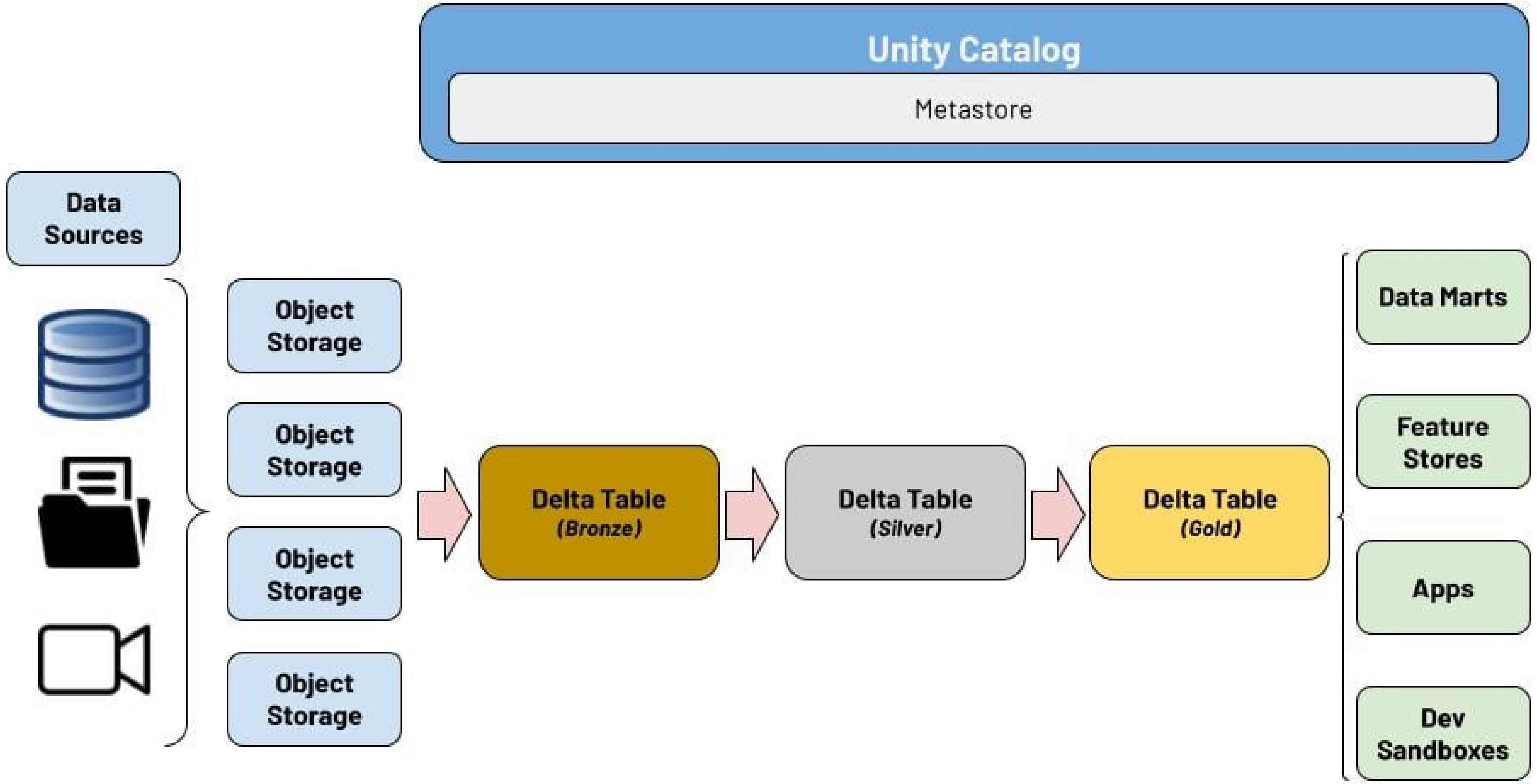
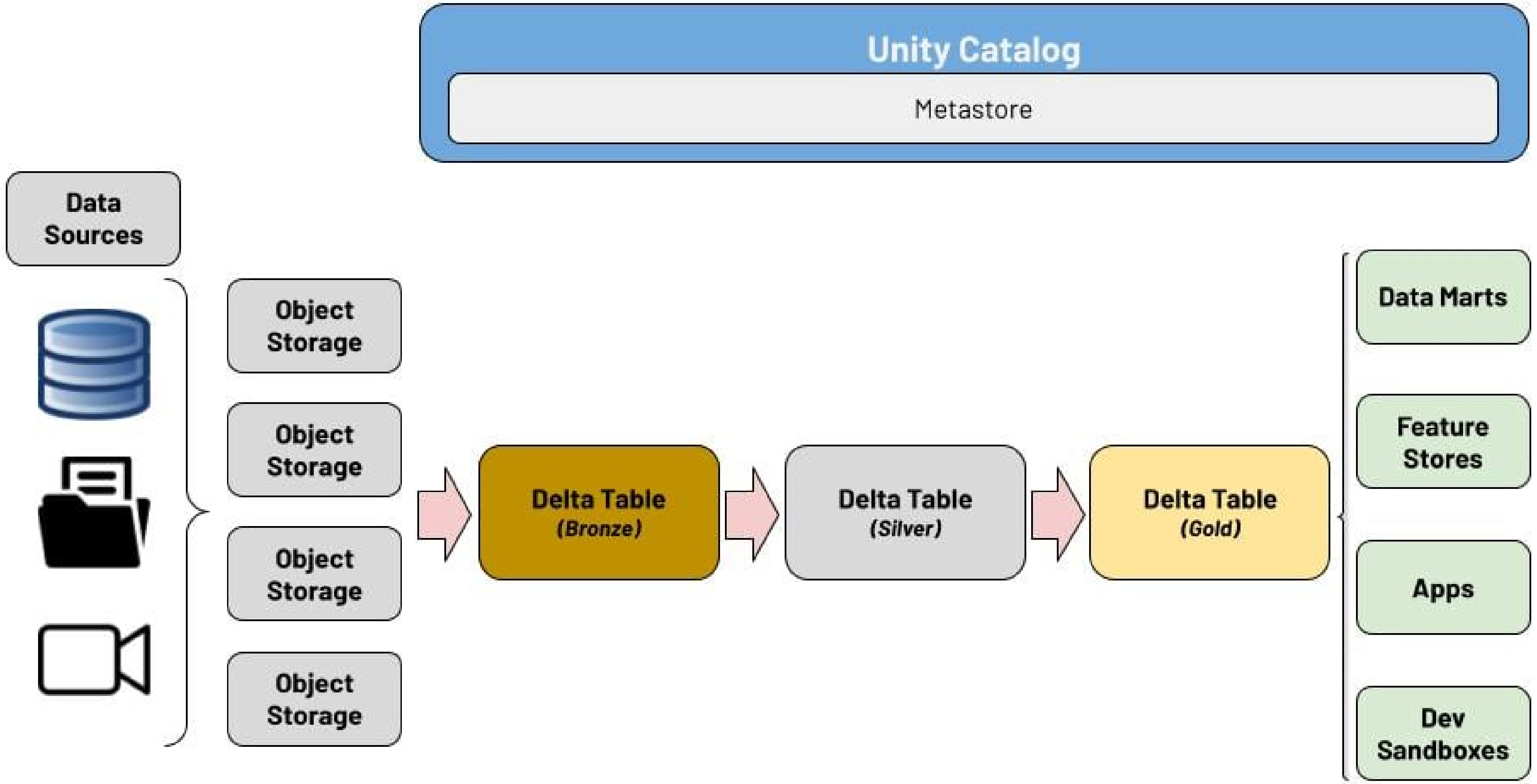
SQL in the lakehouse architecture



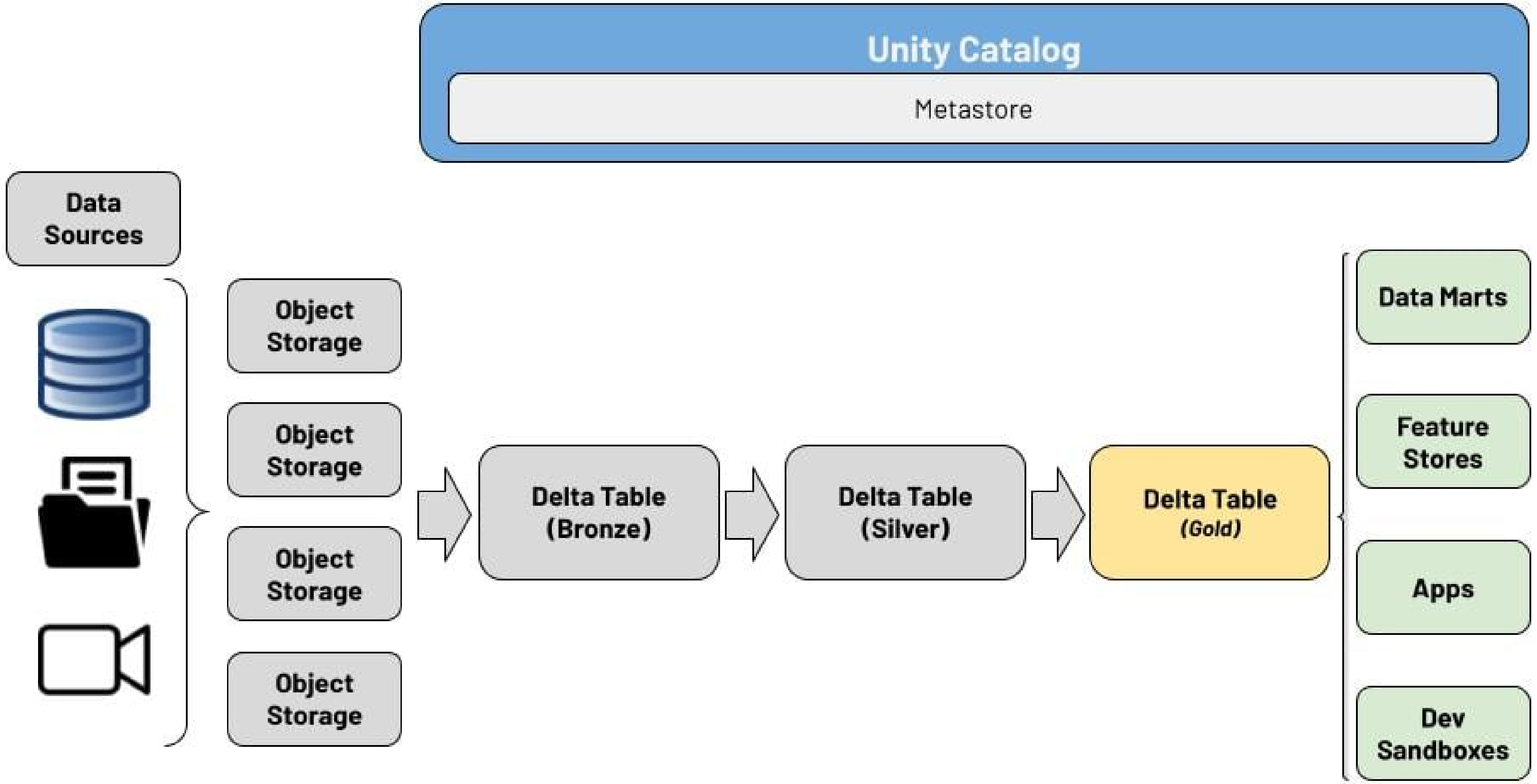
Databricks supports the SQL language throughout the entire platform, and has a first-class experience!

Databricks SQL for data engineering



Databricks SQL can ingest data, support both append and CDC, and transform data through the Medallion Architecture.

Databricks SQL for data analysis



Data analysts can create their queries, visualizations, and dashboards in Databricks natively, or with a partner BI tool.

Thank you!

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